

Unit 3

Communication Protocols

Role of Communication protocols

Embedded system

- Communication protocols are predefined rules that determine how data is transmitted between components in an embedded system.
- Communication protocols in embedded systems are essential for enabling devices to exchange data efficiently and reliably.
- These protocols ensure that devices can communicate without errors, delays, or data loss.

Communication protocols

- Serial communication protocols (I2C, SPI, USB, CAN) enables low-cost, short-distance, and low-power data exchange between internal embedded components like microcontrollers and peripherals.

- Wireless protocols: (Wi-Fi, Bluetooth, ZigBee, LoRa) facilitate long-range, flexible connectivity for IoT devices to cloud systems/sensors.

Topics to be discussed in the Unit

- Modes of Communication:
 - Serial vs Parallel
 - Synchronous vs Asynchronous
- Serial Protocols:
 - CAN, I2C, SPI, USB
- Ethernet (Wired)
- Wireless Protocols
 - ZigBee, Wifi, IrDA, LORA, LOWPAN, MQTT

Choosing a right Protocol

- Exchange information with each other or with cloud server. Not all device can afford high speed and high power. So communication protocols are required.
- A Protocol define the role and method by which data is exchanged.
- A communication protocols may be for short or long range, varying speed.
- Choosing a right Protocol affects
 - Power usage
 - Range
 - Data speed
 - Scalability

Different Serial Communication

Protocols

- Exchange of data between microcontrollers, sensors, and peripherals is based on speed, distance, wiring, and complexity needs.
- No single protocol handles the diverse requirements of embedded systems, ranging from simple sensors to high-speed, robust industrial data transfer.

➤ UART (Universal Asynchronous Receiver/Transmitter):

Simple and easy to implement. It uses only two wires (TX/RX) for asynchronous communication, direct device-to-device connection (point to point connection) e.g connecting a microcontroller to a PC.

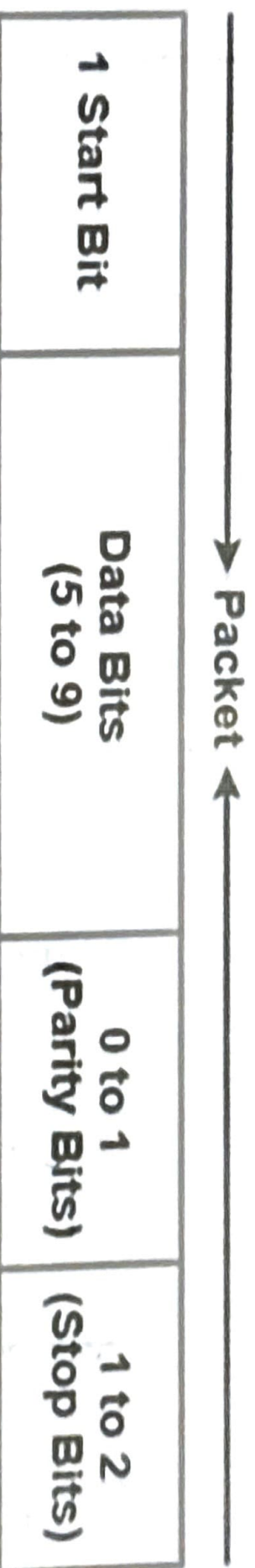
➤ **I2C (Inter-Integrated Circuit):** Required for connecting multiple slave devices (like sensors, EEPROMs) to a single master using only two wires (SDA/SCL). It allows multi-master environment.

➤ **SPI (Serial Peripheral Interface):** Required for high-speed, full-duplex, communication. Used where fast, reliable data transfer is priority.

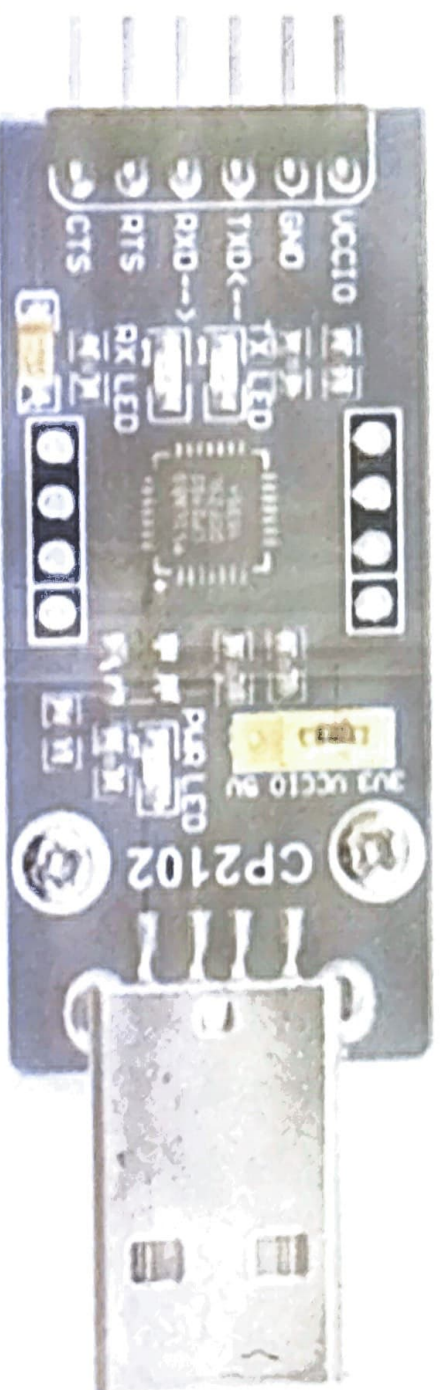
➤ **CAN Bus (Controller Area Network):** Required for industrial and automotive applications where high reliability, error detection, and robust, priority-based messaging over long distances are necessary, even in noisy electrical environments.

➤ **USB (Universal Serial Bus):** Required for high-bandwidth communication between computers and

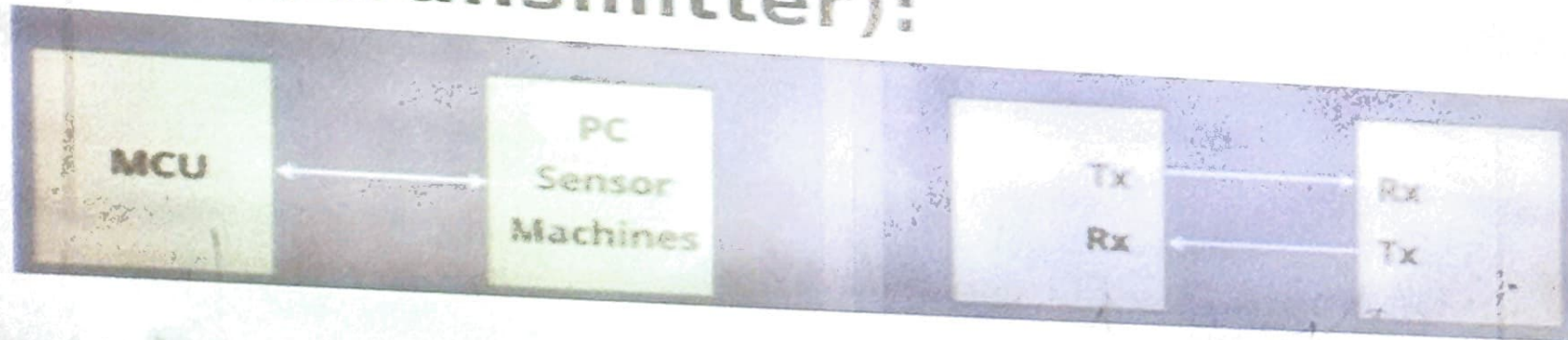
UART Data format



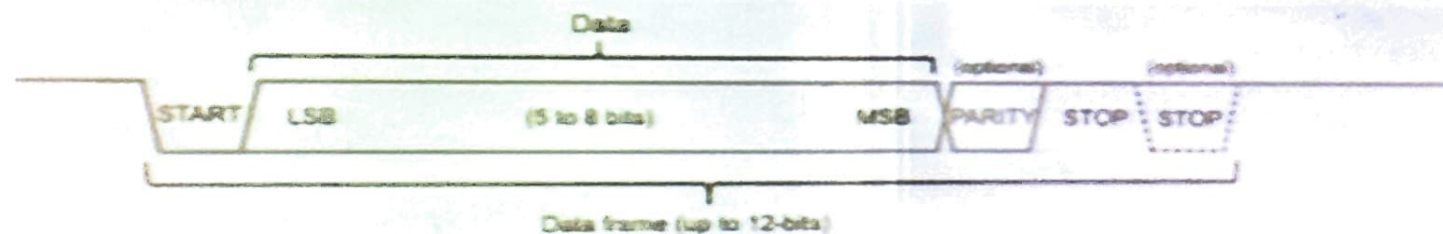
→ Data Frame ←



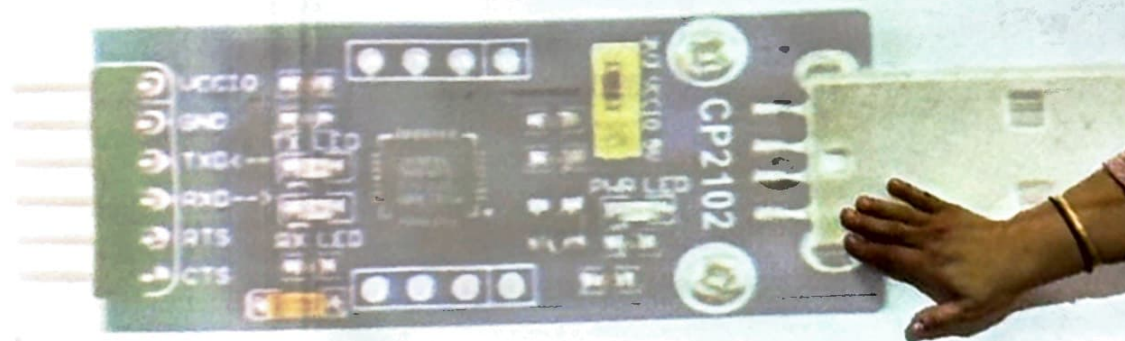
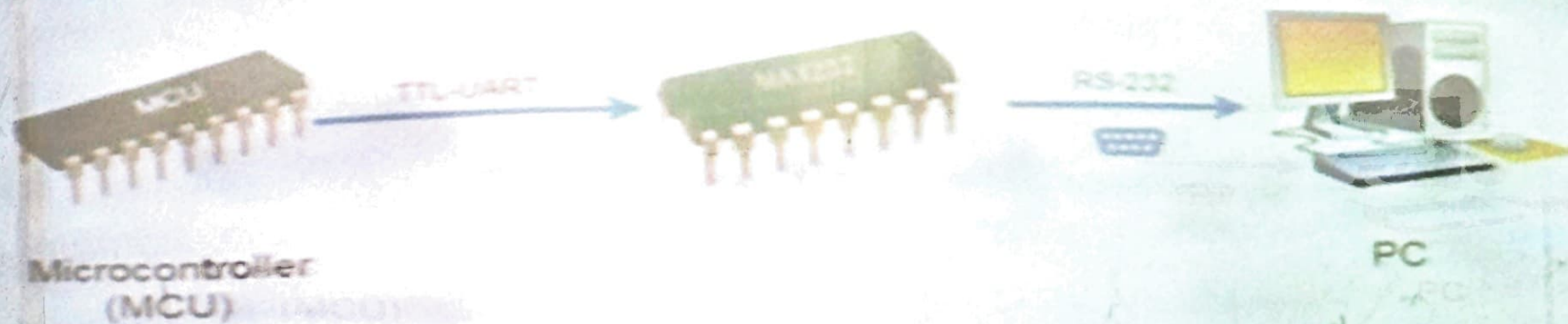
UART (Universal Asynchronous Receiver/Transmitter):



- Data format & Speed is configurable.
Fix the Baud rate (**9600**, **115200**, and **57600** bps)
Fix the data length
- Only 2 wires are required (for Tx & Rx)
- NO clock signal is required, so called asynchronous.
- NRZ line coding is used.
- UART remains valuable for slower-speed and lower-throughput tasks.



UART



Setting Baud Rate in UART

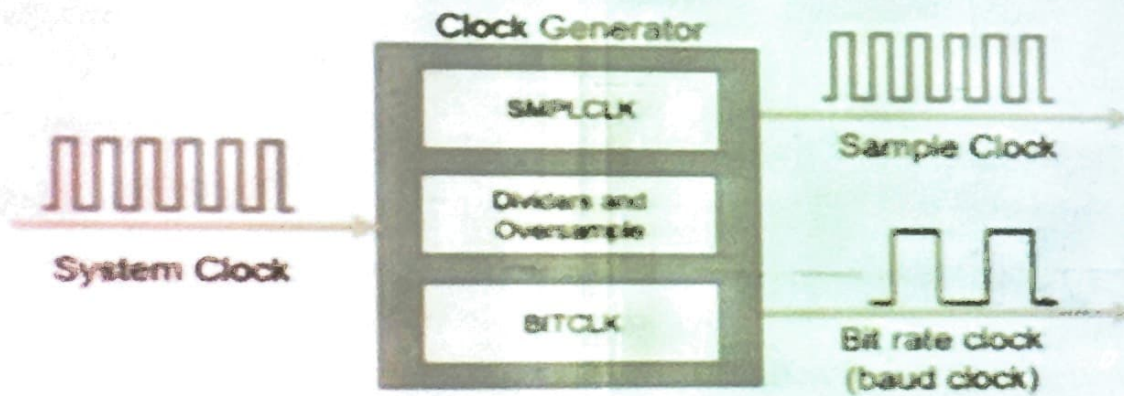
Fclk = 1MHz
BRD = X



9600 baud



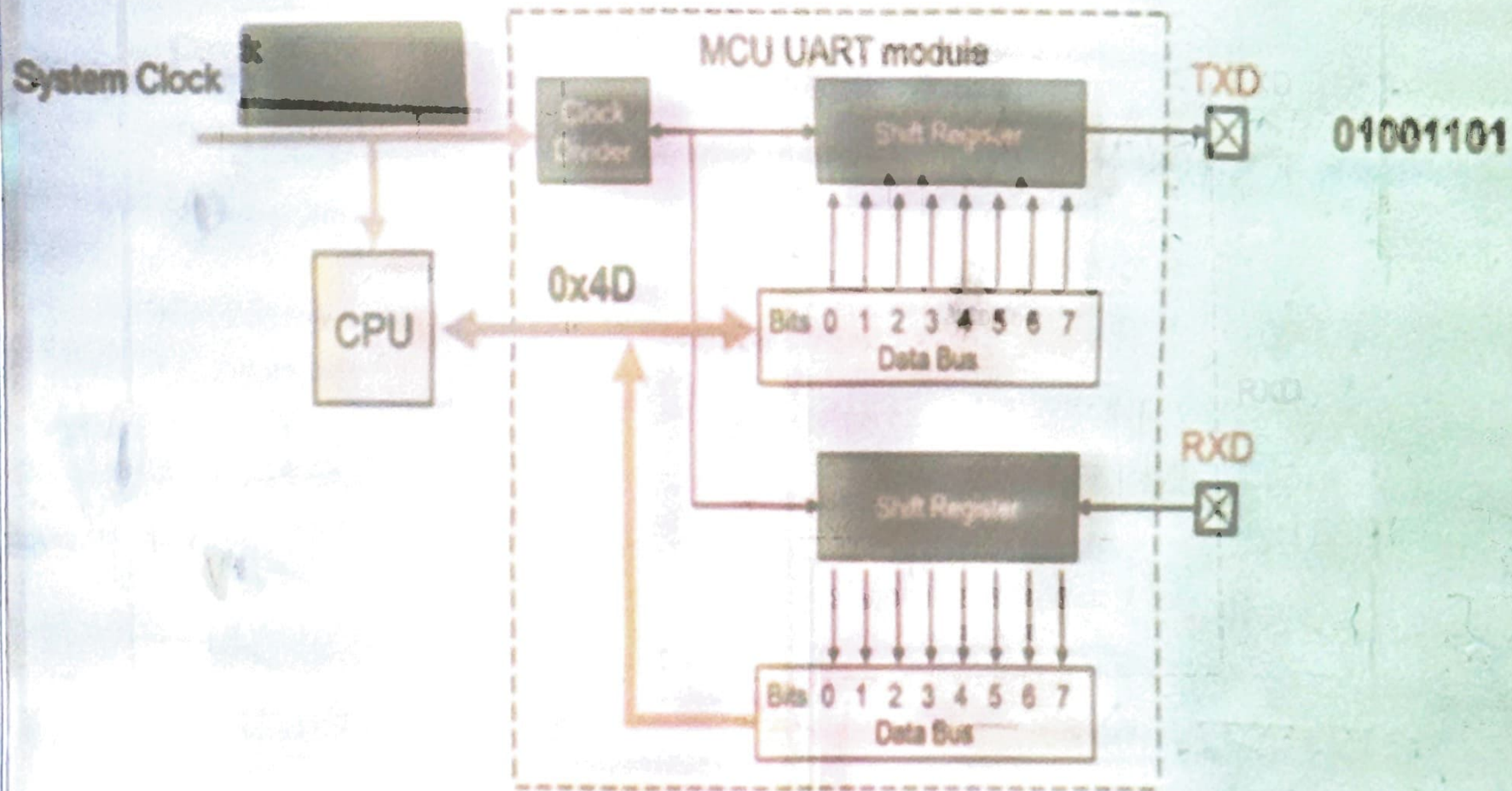
Fclk = 8MHz
BRD = Y



BRD = Baud Rate Divisor



Data Transmission via UART



UART Errors

- Framing Error
- Overflow Error
- Parity Error
- Noise Error
- Break Condition
- Over run error

USB- Universal Serial Bus

- Self configuring. Plug & play
- No control signals required.
- Backward compatible.
- Host swapping (need not to reboot)
- Power charging.
- Small range.
- Single Master slave with Tree topology.
(only host ie PC can initiate transfer)

Version of USB

Version of USB

USB VERSION	RELEASE DATE	NAME	TRANSFER RATES
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USB 1.0	Jan. 1996	Full speed	1.5 Mbps
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USB 1.1	Aug. 1998	Full speed	12 Mbps
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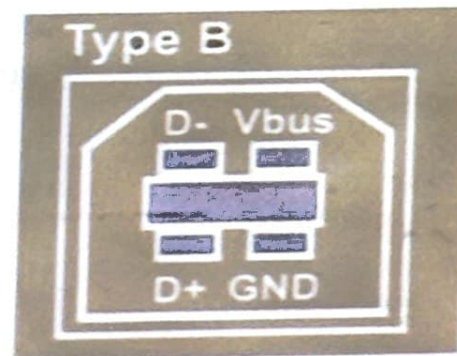
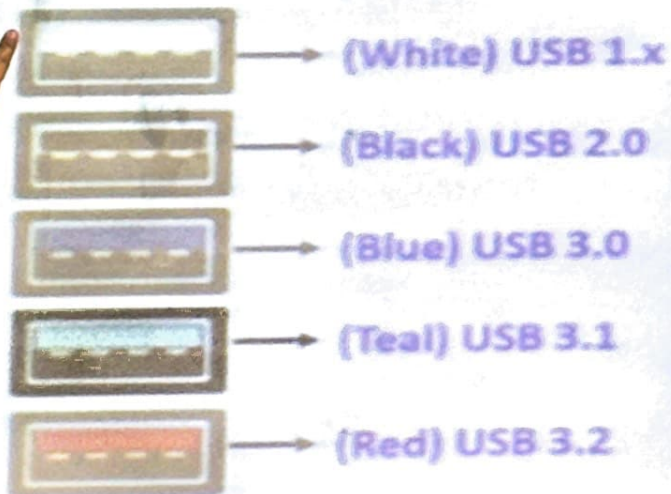
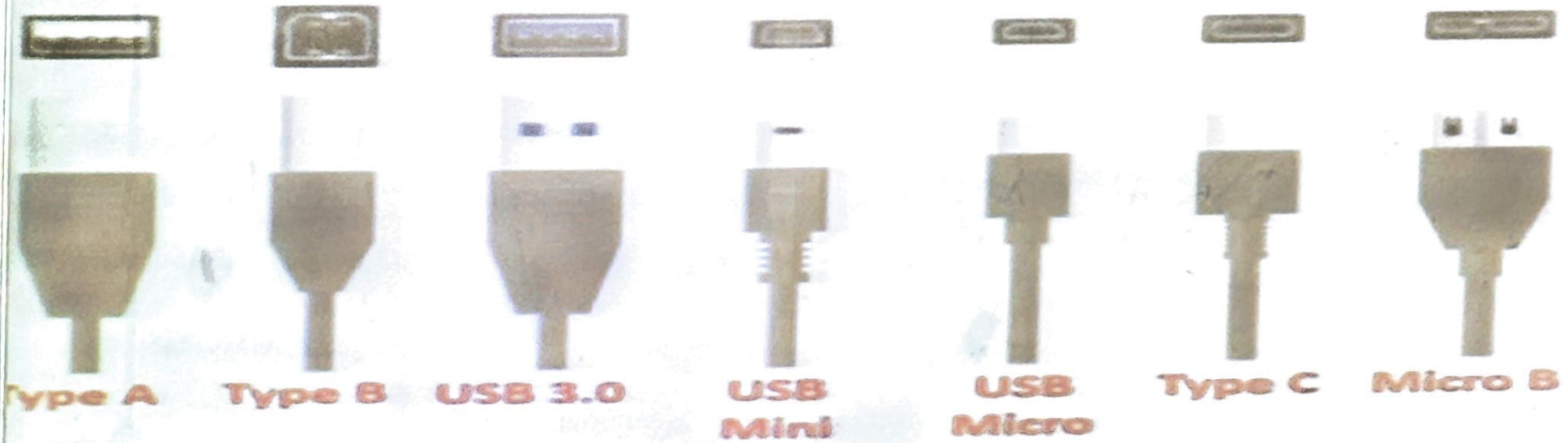
USB 2.0	Apr. 2000	High Speed	480 Mbps
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USB 3.0	Nov. 2008	SuperSpeed	5 Gbps
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USB 3.1	July 2013	SuperSpeed+	10 Gbps
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USB 3.2	Sep. 2017	SuperSpeed+	20 Gbps
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Connectors



Working of USB

Pin	Signal	Color	Description
1	VCC	Orange	+5V
2	D-	White	Data -
3	D+	Green	Data +
4	GND	Black	Ground



and D+ carries
- and D+ carries
fferential voltage
get better SNR.



Working of USB

- USB 2.0 uses a polling mechanism
- only the host can initiate a transaction on the bus.
- Communication in the USB 2.0 protocol is half-duplex.
- it contains a series of frames.
- A transaction is a set of a token packet, a data packet, and a handshake packet

Token Packet :-

Sync	PID	Device Address	End Point	CRC 5-Bit	EOP
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Data Packet:-

Sync	PID	Data	CRC 16-Bit	EOP
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Handshake Packet:-

Sync	PID	EOP
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SOF Packet:-

Sync	PID	Frame number	End Point	CRC	EOP
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USB Architecture

Tree topology

